Paper Simulation II: shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.66	34.48	7.01/7.31/3.95	7.01/7.31/3.95	8.07/8.65/4.82
Scenario II: T1 vs. T0	18.18	47.47	18.88/19.91/14.46	41.42/42.85/36.00	20.96/21.90/16.25
Scenario II: T2 vs. T0	38.34	70.26	38.14/39.71/31.57	58.91/60.86/52.87	41.39/43.21/34.57
Scenario II: T3 vs. T0	60.74	86.05	59.23/61.49/52.87	74.71/76.07/69.50	63.11/65.29/57.04
Scenario II: T4 vs. T0	79.67	95.22	78.14/80.17/72.81	87.07/88.30/83.45	81.25/83.16/76.56
Scenario II: T5 vs. T0	91.79	98.60	90.44/91.33/87.47	94.47/95.41/92.46	92.53/93.49/90.24
Scenario II: T6 vs. T0	97.44	99.73	96.38/97.11/94.94	97.97/98.33/97.07	97.43/97.94/96.49
Scenario II: T7 vs. T0	99.29	99.98	98.89/99.08/98.26	99.44/99.60/98.99	99.32/99.53/98.89
Scenario II: T8 vs. T0	99.94	100.00	99.75/99.82/99.61	99.85/99.91/99.75	99.89/99.92/99.83
Scenario II: T9 vs. T0	99.98	100.00	99.96/99.99/99.91	99.96/99.99/99.93	99.97/100.00/99.95
Scenario II: T10 vs. T0	100.00	100.00	99.96/99.99/99.99	99.97/99.99/99.99	99.99/100.00/99.99
Scenario II: joint power	11.76	39.45	12.35/13.37/8.78	36.71/38.16/31.74	13.89/14.88/10.19

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all ten nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.64	32.61	5.06/5.89/4.24	5.06/5.89/4.24	6.24/7.23/5.25
Scenario II: T1 vs. T0	43.51	75.03	41.36/42.79/37.50	70.88/72.08/68.73	44.87/46.46/41.61
Scenario II: T2 vs. T0	75.37	93.56	72.33/74.07/69.37	88.65/89.27/87.24	75.93/76.97/73.20
Scenario II: T3 vs. T0	93.32	99.23	91.62/92.34/90.32	97.09/97.46/96.65	93.43/94.07/92.48
Scenario II: T4 vs. T0	99.09	99.89	98.78/98.74/98.33	99.60/99.62/99.50	99.24/99.23/98.85
Scenario II: T5 vs. T0	99.92	100.00	99.79/99.83/99.76	99.92/99.97/99.93	99.87/99.96/99.87
Scenario II: T6 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T7 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T8 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T9 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T10 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: joint power	39.10	72.84	36.61/38.55/33.05	69.26/70.54/67.04	40.42/42.08/37.36

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all ten nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: non−shrinkage

N = 40 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	6.42	35.61	7.01/7.31/3.95	7.01/7.31/3.95	8.63/9.11/5.19
Scenario II: T1 vs. T0	19.16	47.97	18.88/19.91/14.46	41.42/42.85/36.00	21.67/22.72/16.75
Scenario II: T2 vs. T0	39.20	69.90	38.14/39.71/31.57	58.91/60.86/52.87	41.73/43.44/35.29
Scenario II: T3 vs. T0	60.80	86.09	59.23/61.49/52.87	74.71/76.07/69.50	63.08/65.39/57.12
Scenario II: T4 vs. T0	79.09	94.95	78.14/80.17/72.81	87.07/88.30/83.45	81.20/82.91/76.36
Scenario II: T5 vs. T0	91.24	98.50	90.44/91.33/87.47	94.47/95.41/92.46	92.25/93.08/89.75
Scenario II: T6 vs. T0	96.91	99.70	96.38/97.11/94.94	97.97/98.33/97.07	97.30/97.81/96.19
Scenario II: T7 vs. T0	99.11	99.96	98.89/99.08/98.26	99.44/99.60/98.99	99.23/99.48/98.79
Scenario II: T8 vs. T0	99.89	100.00	99.75/99.82/99.61	99.85/99.91/99.75	99.86/99.89/99.75
Scenario II: T9 vs. T0	99.98	100.00	99.96/99.99/99.91	99.96/99.99/99.93	99.97/99.99/99.92
Scenario II: T10 vs. T0	100.00	100.00	99.96/99.99/99.99	99.97/99.99/99.99	99.98/99.99/99.99
Scenario II: joint power	12.22	39.53	12.35/13.37/8.78	36.71/38.16/31.74	14.55/15.29/10.64

All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all ten nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Paper Simulation II: non−shrinkage

N = 80 per arm; 30% expected MAR missingness; 40 imputations; 10000 replicates per scenario

Scenario / metric	Naive	Single	Bonferroni PS / PMM / RE	Holm PS / PMM / RE	Proposed PS / PMM / RE
Scenario I: joint FWER	5.55	32.32	5.06/5.89/4.24	5.06/5.89/4.24	6.22/7.18/5.35
Scenario II: T1 vs. T0	44.28	75.31	41.36/42.79/37.50	70.88/72.08/68.73	45.29/47.09/42.01
Scenario II: T2 vs. T0	75.35	93.63	72.33/74.07/69.37	88.65/89.27/87.24	75.62/76.98/73.21
Scenario II: T3 vs. T0	93.06	99.19	91.62/92.34/90.32	97.09/97.46/96.65	93.20/93.86/92.30
Scenario II: T4 vs. T0	98.98	99.90	98.78/98.74/98.33	99.60/99.62/99.50	99.19/99.14/98.76
Scenario II: T5 vs. T0	99.92	100.00	99.79/99.83/99.76	99.92/99.97/99.93	99.86/99.91/99.83
Scenario II: T6 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T7 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T8 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T9 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: T10 vs. T0	100.00	100.00	100.00/100.00/100.00	100.00/100.00/100.00	100.00/100.00/100.00
Scenario II: joint power	39.71	73.11	36.61/38.55/33.05	69.26/70.54/67.04	40.67/42.68/37.68

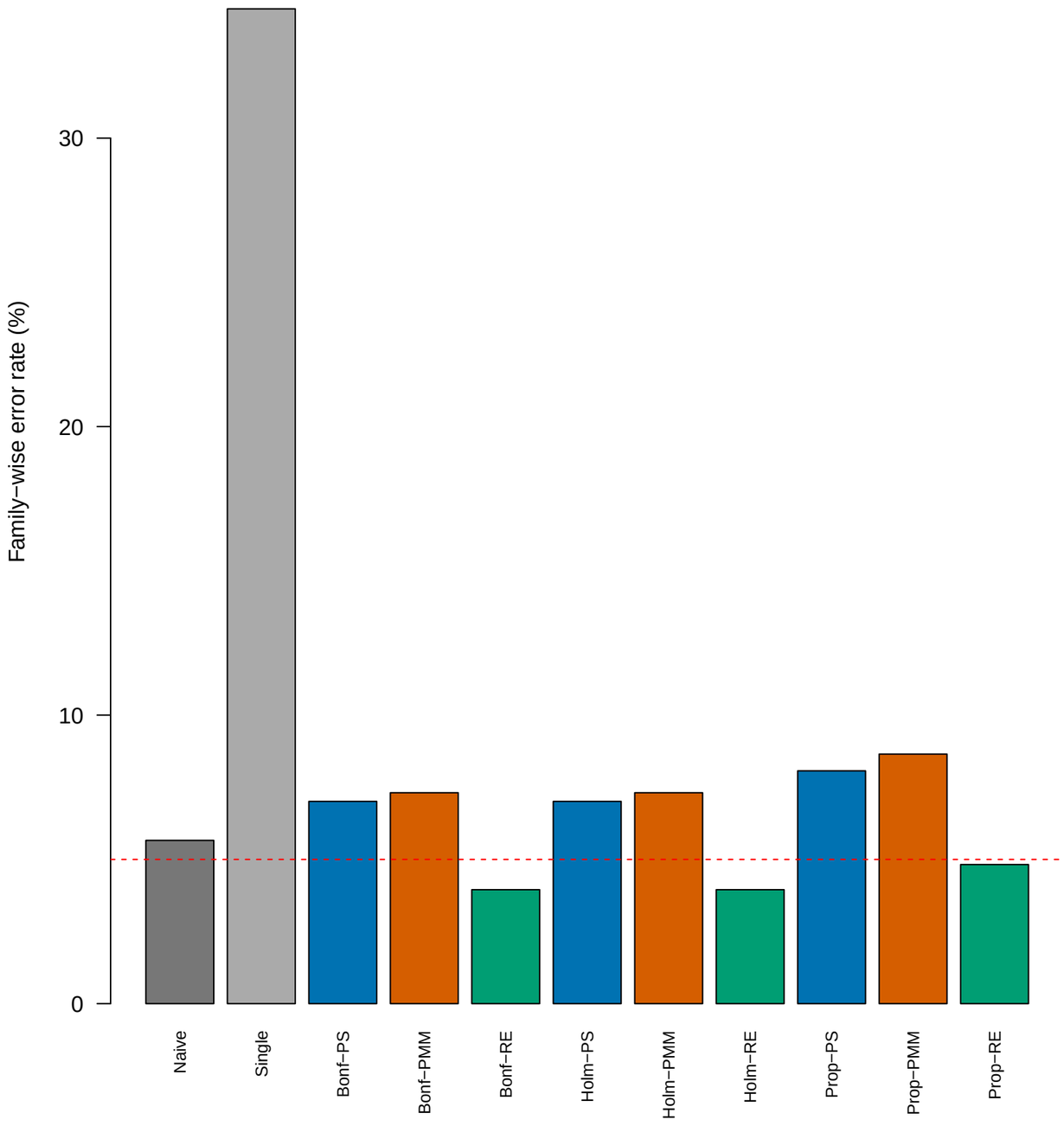
All entries are percentages. PS = propensity−score MI; PMM = predictive mean matching; RE = regression MI.

FWER = reject at least one null in Scenario I. Joint power = reject all ten nulls in Scenario II.

Scenario I : 10000 successful, 0 failed | Scenario II : 10000 successful, 0 failed

Simulation II: shrink − Scenario I

N = 40



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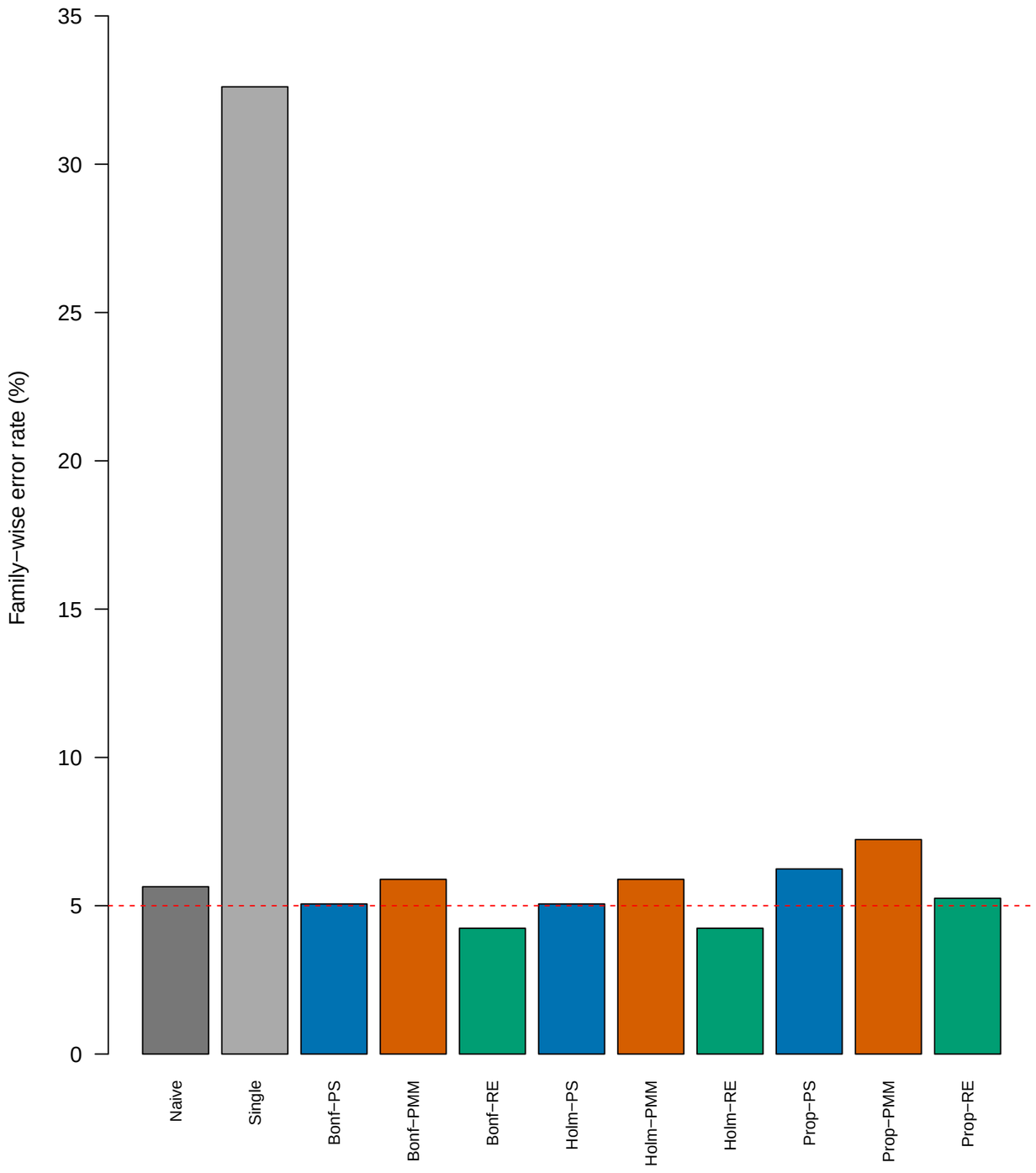
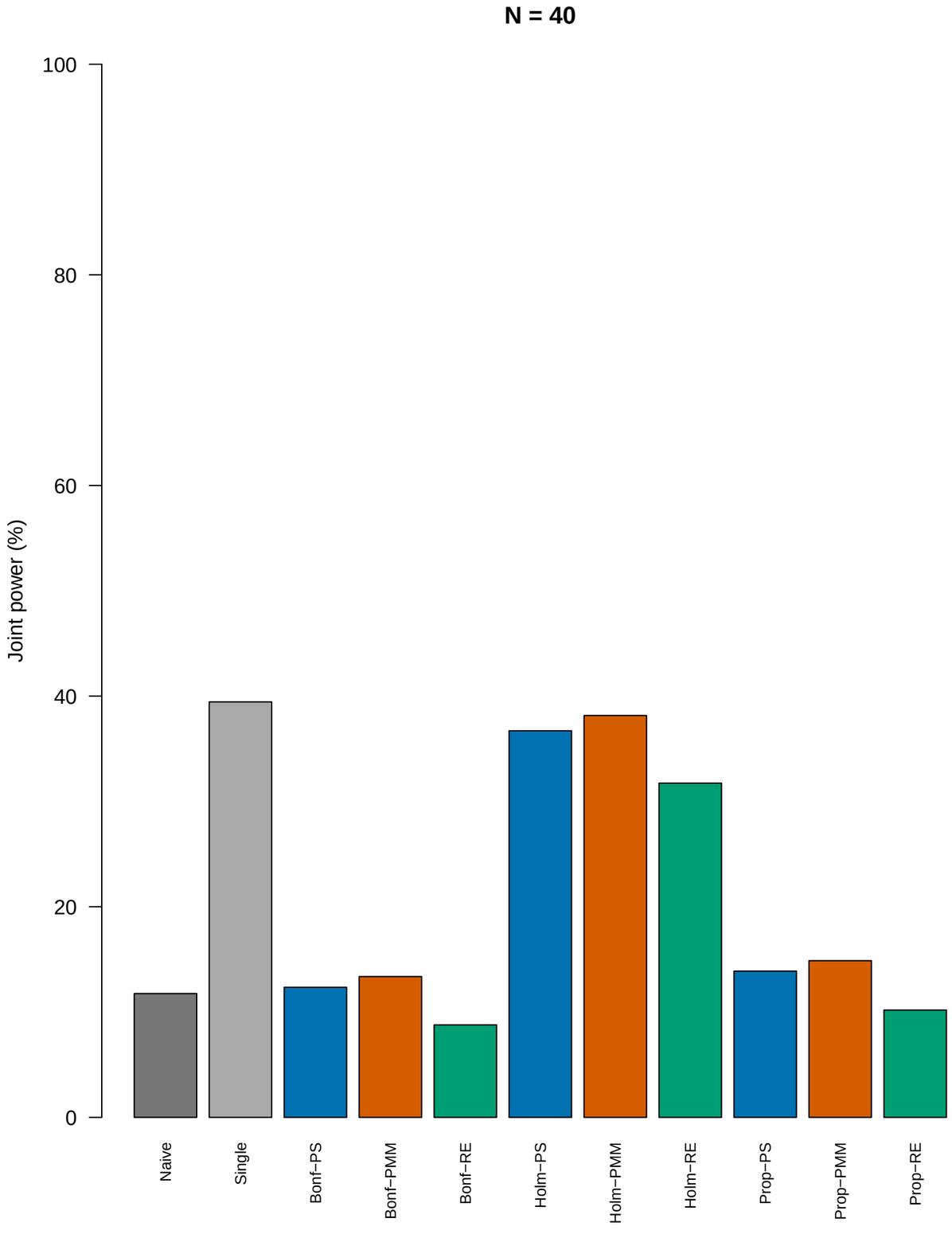
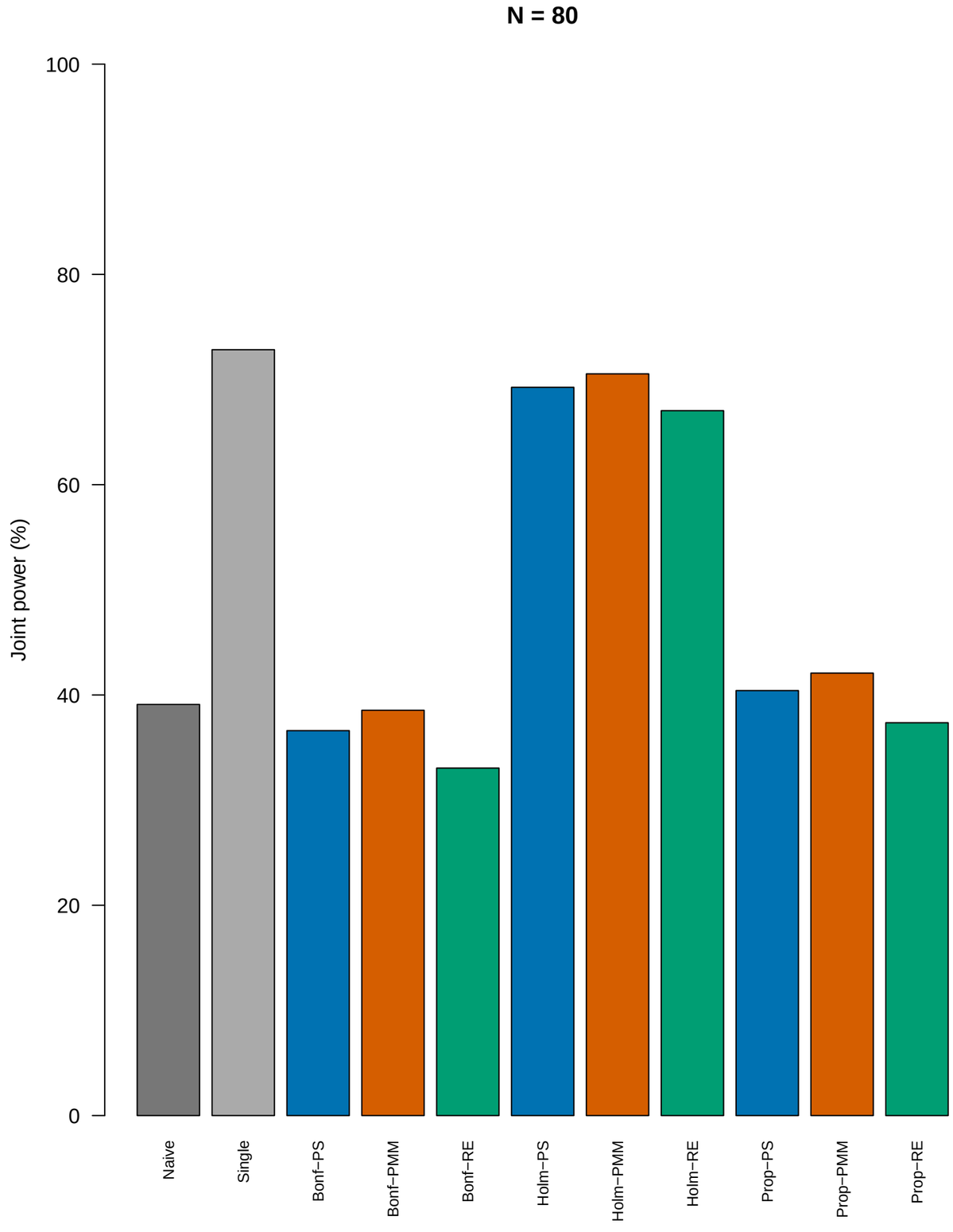
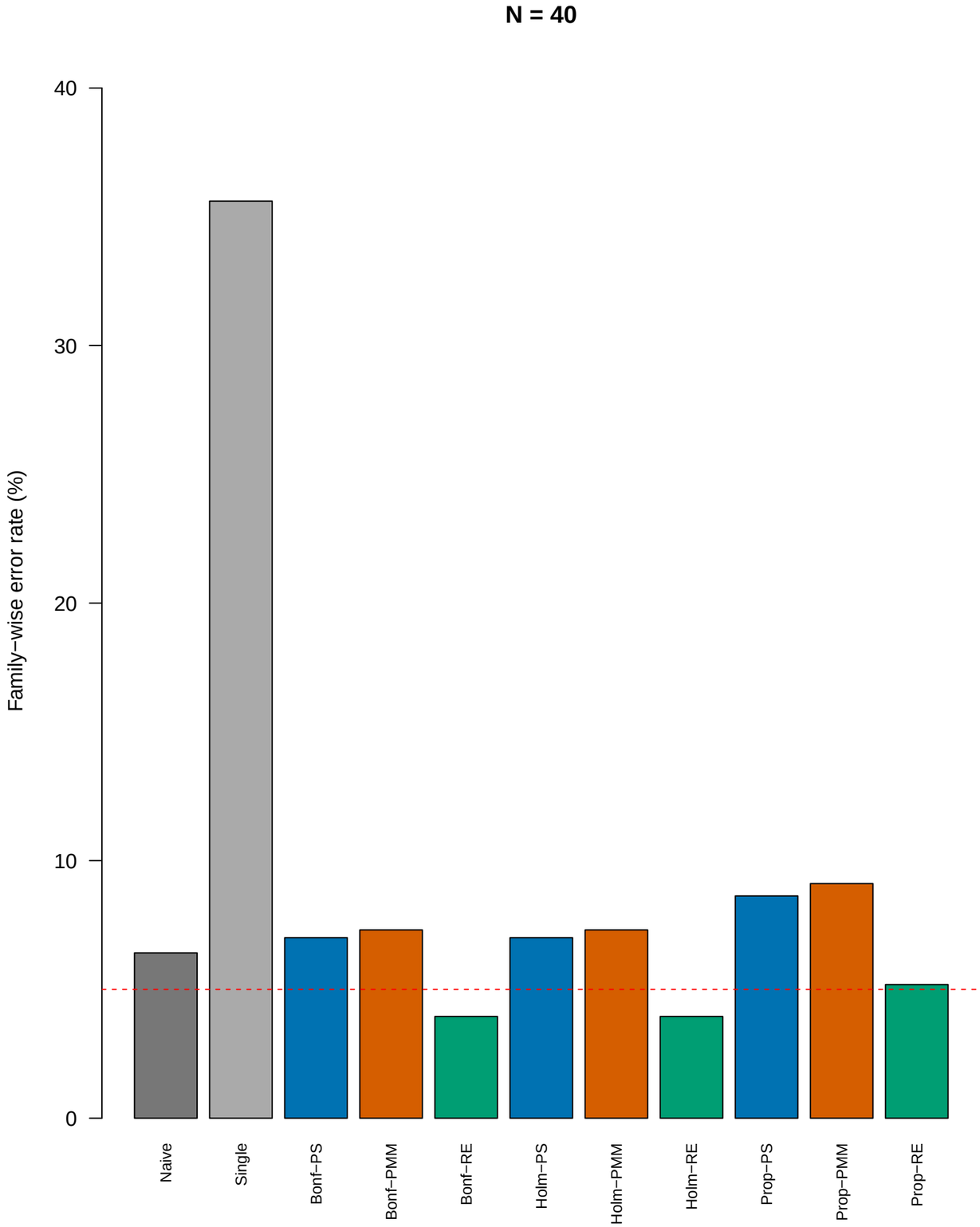


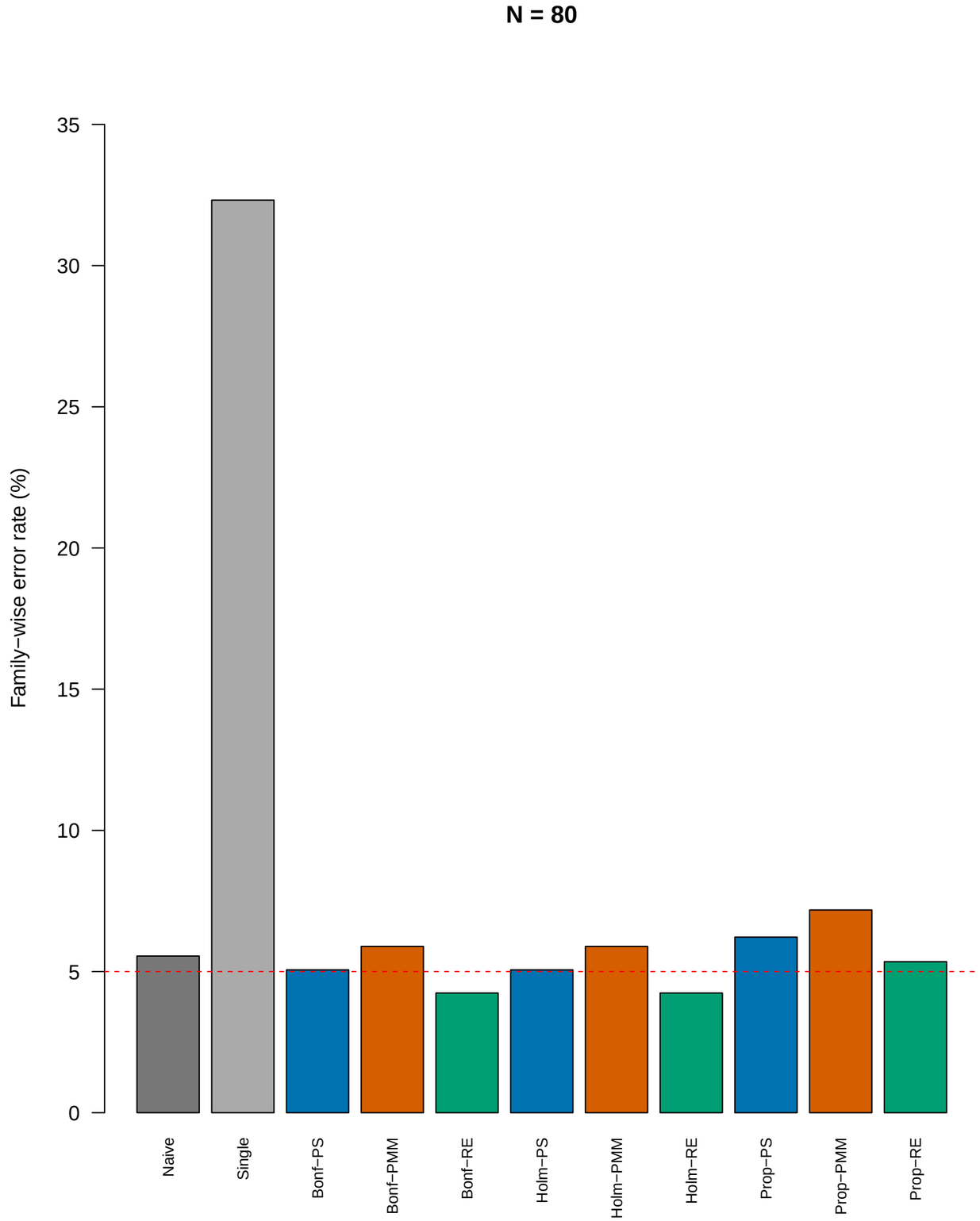
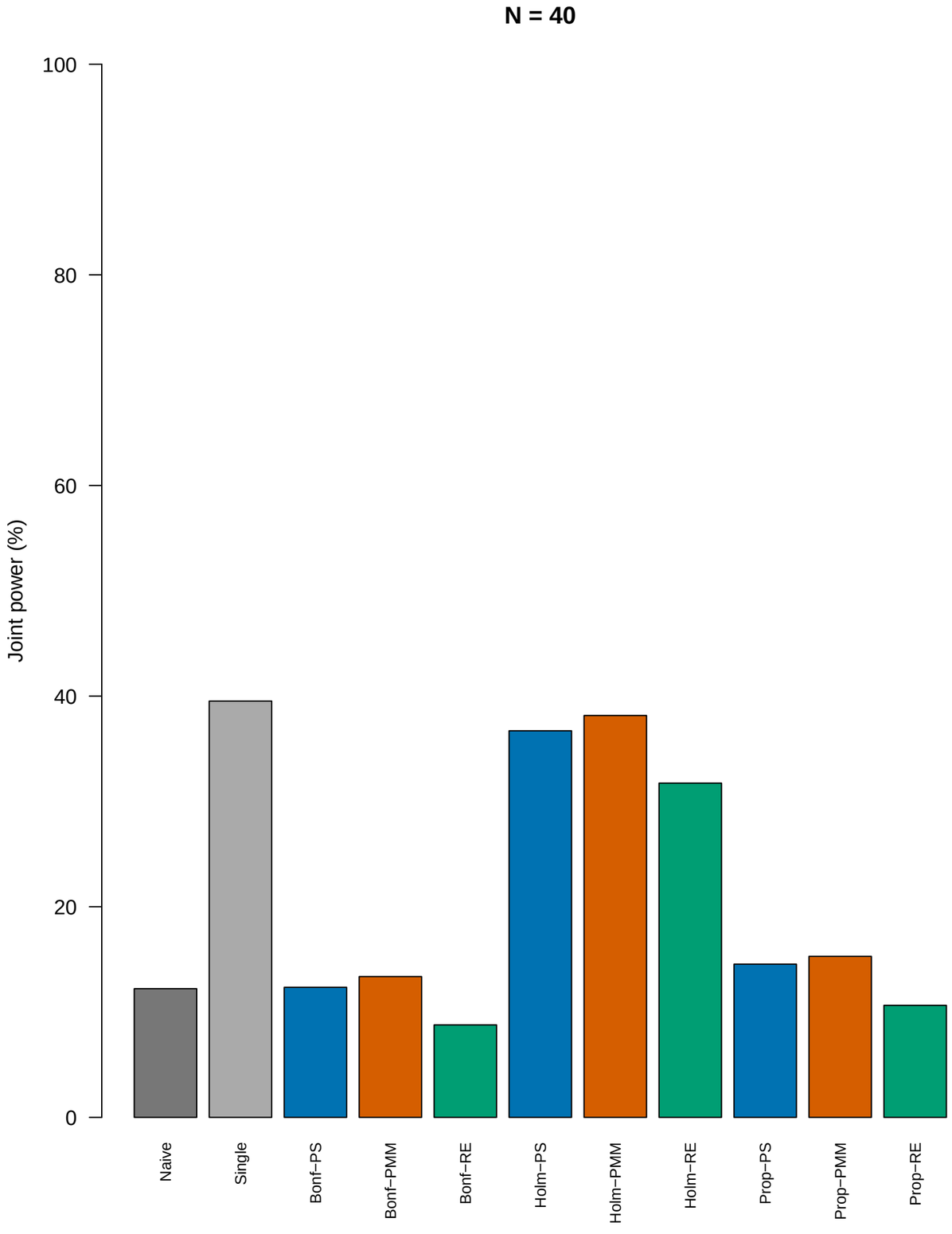
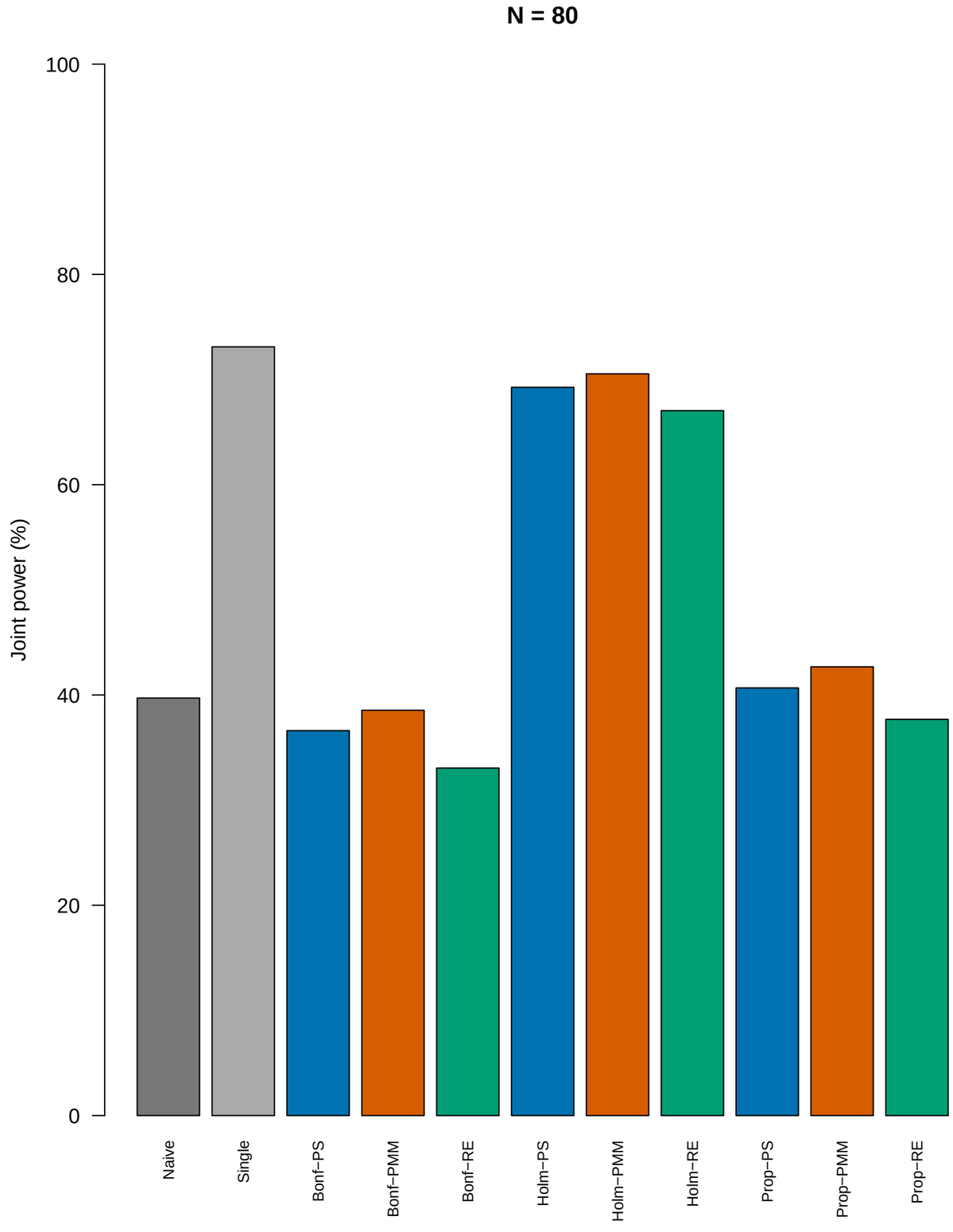
Simulation II: shrink − Scenario II





Simulation II: nonshrink − Scenario I





Simulation II: nonshrink − Scenario II